

# WELL COMPLETION REPORT (SYNTHETIC)

## OIL-NHK-006 - Nahorkatiya-006

NWIS prototype document - not an Oil India Limited operational record

|                         |                                         |
|-------------------------|-----------------------------------------|
| API Number              | OIL-ASM-000062                          |
| Field / District        | Nahorkatiya / Dibrugarh                 |
| Block                   | PEL-ASSAM-02                            |
| Basin                   | Assam-Arakan                            |
| Formation               | Tipam (TIP)                             |
| Spud / Completion       | 1999-03-30 / 1999-06-03                 |
| Measured / TVD          | 2660 m / 2617 m                         |
| Trajectory              | Directional; inc 28.3 deg; az 274.4 deg |
| Formation Interval      | 1160-1977 m                             |
| Lithology               | Fluvial sandstone with clay streaks     |
| Pore Pressure Reference | 2423 psi                                |
| Fracture Gradient       | 14.54 ppg                               |
| Mud System              | Low-Solids WBM                          |
| NWIS Well Risk          | Moderate (50.8/100)                     |

### Casing Program

| String            | Set Depth (m) | Purpose      |
|-------------------|---------------|--------------|
| 20 in conductor   | 80            | Conductor    |
| 13 3/8 in casing  | 665           | Surface      |
| 9 5/8 in casing   | 1542          | Intermediate |
| 7 in liner/casing | 2340          | Production   |

### Significant Historical Events and Lessons

| Date / Depth         | Event           | Severity | Evidence / mitigation / lesson                                                                                                                                                                                  |
|----------------------|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1999-05-05<br>1580 m | Stuck Pipe      | Critical | High drag was recorded while circulating remained possible. Mitigation: Condition mud and verify cuttings transport before connection. Lesson: Avoid prolonged static periods in high-drag interval.            |
| 1999-04-19<br>944 m  | Torque Spike    | High     | High torque developed while WOB remained elevated. Mitigation: Reduce WOB/RPM and circulate before resuming normal parameters. Lesson: Use offset torque trend to set adaptive operating envelope.              |
| 1999-04-14<br>759 m  | Mud Loss        | Medium   | Active pit volume declined during stable pump rate. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Track flow balance and pit volume trend before losses become severe. |
| 1999-04-20<br>1006 m | Stuck Pipe      | Medium   | Torque and drag increased while ROP declined. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Avoid prolonged static periods in high-drag interval.                         |
| 1999-04-24<br>1166 m | Mud Loss        | Medium   | Active pit volume declined during stable pump rate. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.           |
| 1999-04-26<br>1226 m | Cementing Issue | Medium   | Displacement pressure became unstable. Mitigation: Use staged cementing response when returns deteriorate. Lesson: Plan conservative displacement rate in weak formations.                                      |
| 1999-05-12<br>1831 m | Cementing Issue | Medium   | Displacement pressure became unstable. Mitigation: Control displacement rate and reconcile pumped/returned volumes. Lesson: Plan conservative displacement rate in weak formations.                             |

### Completion Summary

The synthetic well reached 2660 m MD in the Tipam interval. The NWIS knowledge record links 16 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.